

Name: _____

Maths Class Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Extension 2 Mathematics

HSC Assessment Task 2

June 2012

General Instructions

- Working time – 70 minutes
- Write using **black or blue pen**
- Board-approved calculators may be used
- **All necessary working** should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Start each question on a **new page**.
- Place your papers **in order** with the question paper on top and staple or pin them.

Total Marks - 52

- Attempt Questions 1 – 4
- Mark values are shown with the questions.

Question 1**13 Marks**

- (a) Find $\int x \cos(x^2) dx$ 2
- (b) Find $\int \frac{dx}{\sqrt{5-4x-x^2}}$ 2
- (c) Use partial fractions to find $\int \frac{-2dx}{x^2+3x-4}$ 3
- (d) Using integration by parts, evaluate $\int_1^e \ln x dx$ 2
- (e) Using the substitution $x = 2\sin\theta$ or otherwise, evaluate $\int_0^1 \sqrt{4-x^2} dx$ 4
leaving your answer in exact form.

Question 2**13 Marks**

- (a) Find $\int \frac{x^2+2x-3}{x+1} dx$ 3
- (b) Find $\int \frac{x-1}{\sqrt{x+1}} dx$ 2
- (c) Consider the rectangular hyperbola, $R: xy = c^2$
- (i) Find the foci. 2
- (ii) Write the equations of the directrices. 1
- (iii) Find the equation of the tangent to R at $P(ct, \frac{c}{t})$. 2
- (iv) This tangent cuts the coordinate axes at A and B . Prove that $PA = PB$. 3

(Questions 3 and 4 on reverse of page)

Question 3

13 Marks

(a) If $f(x) = x - \frac{1}{x}$, provide separate half page sketches of the graphs of the following:

- | | | |
|-------|----------------------|---|
| (i) | $y = f(x)$ | 2 |
| (ii) | $y = \sqrt{f(x)}$ | 2 |
| (iii) | $y = \frac{1}{f(x)}$ | 2 |
| (iv) | $y = f'(x)$ | 2 |
| (v) | $y = f(x)$ | 2 |

(b) Solve the equation $4x^3 - 8x^2 + 5x - 1 = 0$ given that it has a double root. 3

Question 4

13 Marks

(a) If $2x^3 - 4x^2 + 6x - 1 = 0$ has roots α , β and γ , find:

- | | | |
|-------|---|---|
| (i) | $\alpha^3 + \beta^3 + \gamma^3$ | 3 |
| (ii) | $\alpha^4 + \beta^4 + \gamma^4$ | 2 |
| (iii) | $\alpha^2\beta + \alpha^2\gamma + \beta^2\gamma + \beta^2\alpha + \gamma^2\alpha + \gamma^2\beta$ | 2 |

(b) $1 - i$ and $2 + i$ are zeroes of a monic polynomial, $P(x)$, with real coefficients and degree 4.

- | | | |
|------|--|---|
| (i) | Express $P(x)$ as a product of two real quadratic factors. | 2 |
| (ii) | Explain briefly why the polynomial $P(x)$ cannot take negative values. | 1 |

(c) The equation $x^3 - 6x^2 + 7x - 3 = 0$ has roots α , β and γ .

- | | | |
|------|--|---|
| (i) | Write an equation which has the roots α^2 , β^2 and γ^2 . | 2 |
| (ii) | It is known that the solution to a given problem is the average of the roots of the equation $x^3 - 6x^2 + 7x - 3 = 0$. | 1 |

Without finding the roots, determine the solution to the problem.

End of Exam

HSC 2012 Ass 2 Extension 2 - June SOLUTIONS

Q1 a) $\int x \cos(x^2) dx = \frac{1}{2} \int 2x \cos(x^2) dx$
 $= \frac{1}{2} \sin(x^2) + C$

b) $\int \frac{dx}{\sqrt{5-4x-x^2}} = \int \frac{dx}{\sqrt{9-(x+2)^2}}$
 $= \sin^{-1}\left(\frac{x+2}{3}\right) + C$

c) $\int \frac{-2 dx}{x^2+3x-4} = \int \frac{-2 dx}{(x+4)(x-1)}$
 $= \int \frac{A}{x+4} + \frac{B}{x-1} dx$ $A(x-1) + B(x+4) = -2$
 $\text{let } x=1, \quad 5B = -2$
 $\therefore B = -\frac{2}{5}$
 $\text{let } x=-4, \quad -5A = -2$
 $\therefore A = \frac{2}{5}$
 $= \frac{2}{5} \int \frac{dx}{x+4} - \frac{2}{5} \int \frac{dx}{x-1}$
 $= \frac{2}{5} \ln(x+4) - \frac{2}{5} \ln(x-1) + C$
 $= \frac{2}{5} \ln \frac{(x+4)}{(x-1)} + C$

d) $\int_1^e \ln x dx = [x \ln x]_1^e - \int_1^e x \cdot \frac{1}{x} dx$ $\text{let } u = \ln x$
 $du = \frac{1}{x} dx$
 $= e - [x]_1^e$ $\text{let } v = x$
 $dv = dx$
 $= e - (e-1)$
 $= 1$

e) $\int_0^1 \sqrt{4-x^2} dx =$ $\text{let } x = 2 \sin \theta$
 $\therefore dx = 2 \cos \theta d\theta$
 $\text{When } x=1, \quad \sin \theta = \frac{1}{2}$
 $\therefore \theta = \frac{\pi}{6}$
 $\text{When } x=0, \quad \theta = 0$
 $= \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} 2 \cos \theta d\theta$
 $= \int_0^{\frac{\pi}{6}} 4 \cos^2 \theta d\theta$
 $= \int_0^{\frac{\pi}{6}} 2(\cos 2\theta + 1) d\theta$
 $= 2 \left[\frac{\sin 2\theta}{2} + \theta \right]_0^{\frac{\pi}{6}}$
 $= 2 \left[\left(\frac{\sqrt{3}}{4} + \frac{\pi}{6} \right) - (0+0) \right]$
 $= \frac{\sqrt{3}}{2} + \frac{\pi}{3}$

Q2 a) $\int \frac{x^2 + 2x - 3}{x+1} dx$

$$= \int \left((x+1) - \frac{4}{x+1} \right) dx$$

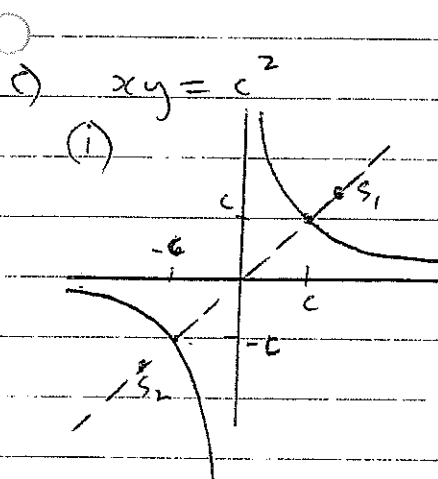
$$= \frac{x^2}{2} + x - 4 \ln(x+1) + C$$

$$\begin{array}{r} x+1 \\ x+1 \overline{) x^2 + 2x - 3} \\ \underline{x^2 + x} \\ x - 3 \\ \underline{x+1} \\ -4 \end{array}$$

b) $\int \frac{x-1}{\sqrt{x+1}} dx = \int \frac{x+1}{\sqrt{x+1}} dx - \int \frac{2}{\sqrt{x+1}} dx$

$$= \int \sqrt{x+1} - \frac{2}{\sqrt{x+1}} dx$$

$$= \frac{2}{3} (\sqrt{x+1})^{\frac{3}{2}} - 4 \sqrt{x+1} + C$$



$S_1 = \sqrt{2}c, \sqrt{2}c$
 $S_2 = -\sqrt{2}c, -\sqrt{2}c$

(ii) directrices $\Rightarrow x+y = \pm\sqrt{2}c$

(iv) When $x=0$, $t^2y = 2ct$

$\therefore y = \frac{2c}{t}$ $A(0, \frac{2c}{t})$

When $y=0$, $x = 2tc$

$\therefore B(2tc, 0)$

Midpoint AB = $(\frac{0+2tc}{2}, \frac{\frac{2c}{t}+0}{2})$

$= (tc, \frac{c}{t})$
 $= P$

$\therefore PA = PB$

(iii) Tangent to R at $(ct, \frac{c}{t})$

$y = \frac{c^2}{x}$
 $\therefore \frac{dy}{dx} = -\frac{c^2}{x^2}$

$= -\frac{c^2}{c^2+t^2}$ when $x=ct$
 $= -\frac{1}{t^2}$

$\therefore$ Equation of tangent is...

$y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$

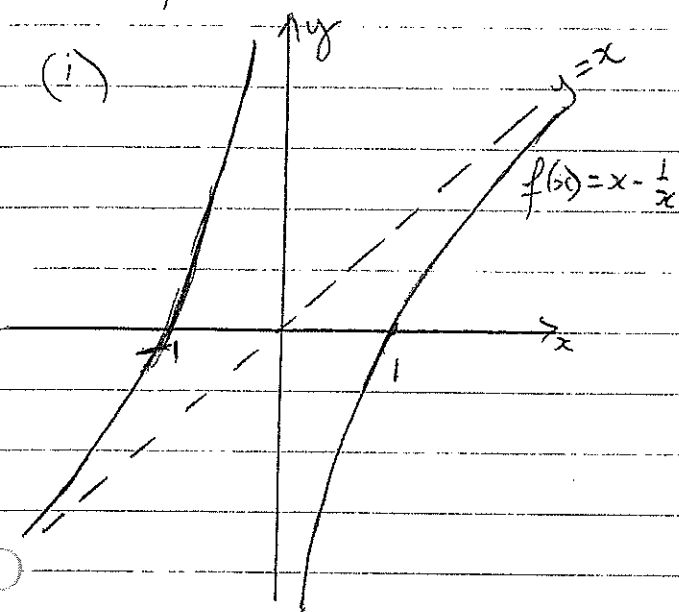
$t^2y - tc = -x + ct$

$\therefore x + t^2y = 2ct$

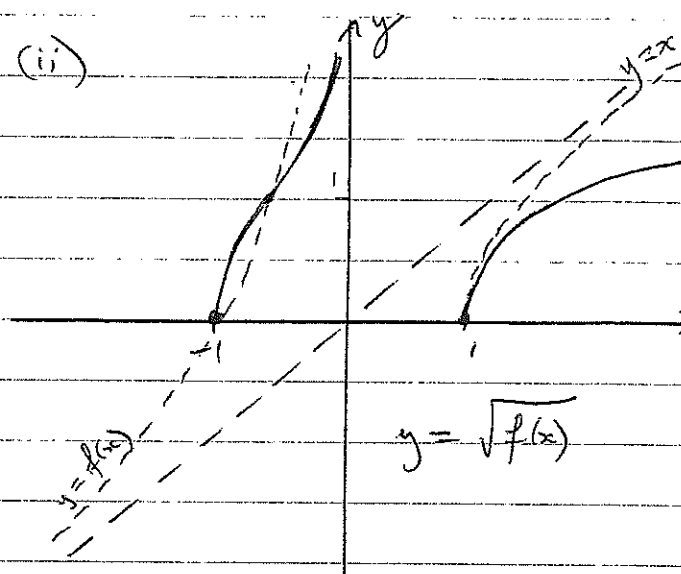
Q3

a) $f(x) = x - \frac{1}{x}$

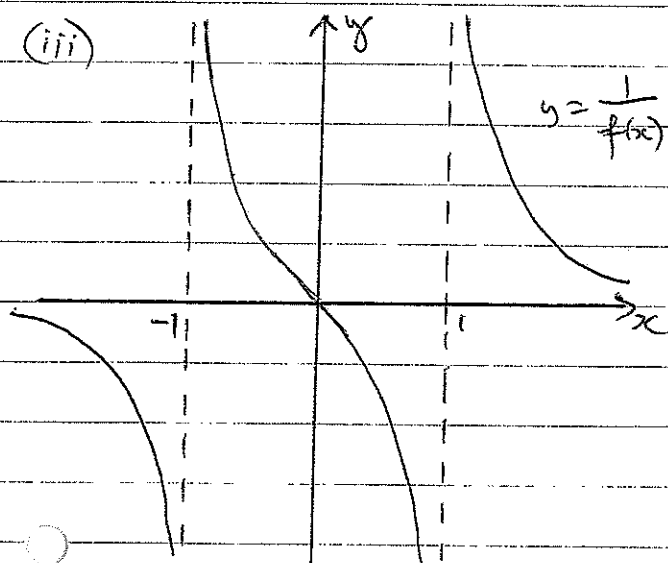
(i)



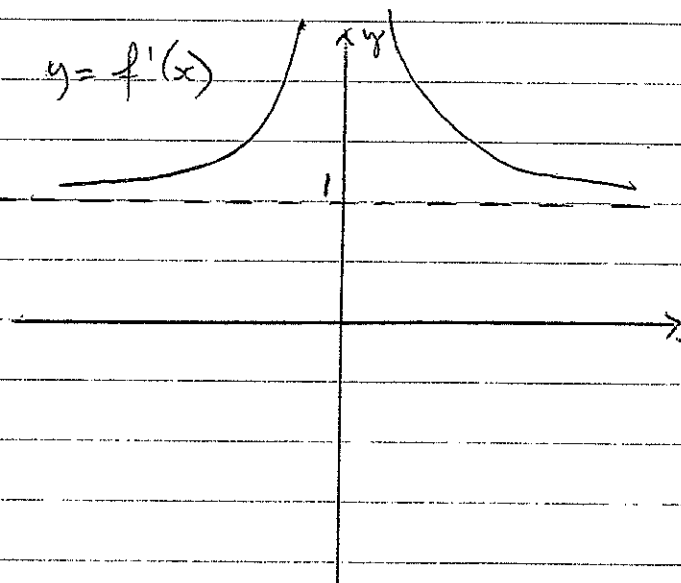
(ii)



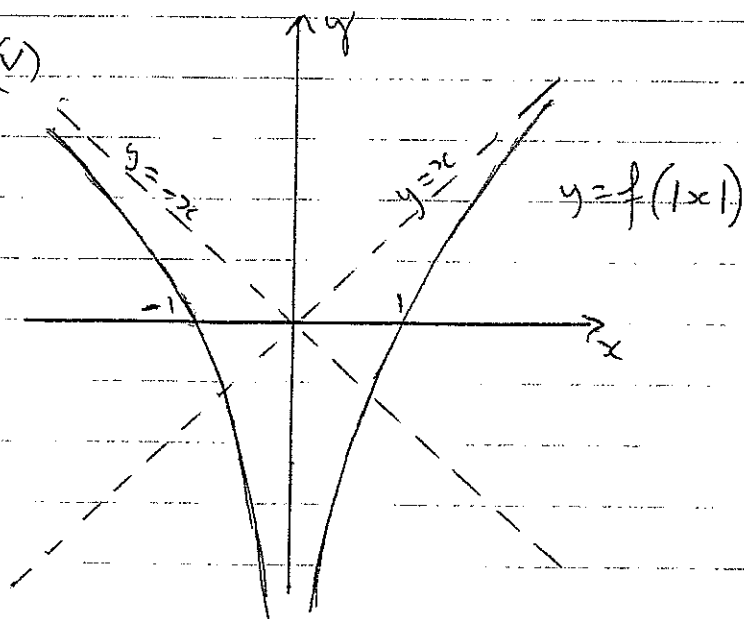
(iii)



(iv) $y = f'(x)$



(v)



b) Let $P(x) = 4x^3 - 8x^2 + 5x - 1$

$\therefore P'(x) = 12x^2 - 16x + 5$

$= (2x-1)(6x-5)$

$= 0$ when $x = \frac{1}{2}, \frac{5}{6}$

Check $x = \frac{1}{2}$ for double root of $P(x)$

$P\left(\frac{1}{2}\right) = \frac{1}{2} - 2 + \frac{5}{2} - 1$
 $= 0$

Product of roots of $P(x) = \frac{-a}{q}$
 $= \frac{1}{4}$

$\therefore$ Other root is 1 as $\frac{1}{2} \times \frac{1}{2} \times 1 = \frac{1}{4}$

$\therefore$ Roots of $P(x)$ are $x = \frac{1}{2}, \frac{1}{2}, 1$

Q4

a) Let $2x^3 - 4x^2 + 6x - 1 = 0$ have roots α, β and γ .

$$\sum \alpha = -\frac{b}{a} = 2$$

$$\sum \alpha\beta = \frac{c}{a} = 3$$

$$\alpha\beta\gamma = -\frac{d}{a} = \frac{1}{2}$$

$$\begin{aligned} \text{(i) Now } \sum \alpha^2 &= (\sum \alpha)^2 - 2\sum \alpha\beta \\ &= 4 - 6 \\ &= -2 \end{aligned}$$

(ii) Multiplying by x and adding again

$$2\sum \alpha^4 - 4\sum \alpha^3 + 6\sum \alpha^2 - \sum \alpha = 0$$

$$\therefore 2\sum \alpha^4 + 34 - 12 - 2 = 0$$

$$\therefore 2\sum \alpha^4 = -20$$

$$\therefore \sum \alpha^4 = -10$$

Also as α, β and γ are roots...

$$2\alpha^3 - 4\alpha^2 + 6\alpha - 1 = 0$$

$$2\beta^3 - 4\beta^2 + 6\beta - 1 = 0$$

$$2\gamma^3 - 4\gamma^2 + 6\gamma - 1 = 0$$

$$\text{By addition, } 2\sum \alpha^3 - 4\sum \alpha^2 + 6\sum \alpha - 3 = 0$$

$$\therefore 2\sum \alpha^3 + 8 + 12 - 3 = 0$$

$$\therefore \sum \alpha^3 = -17$$

$$\therefore \sum \alpha^3 = \frac{-17}{2}$$

(iii) Now, $(\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma)$

$$\begin{aligned} &= (\alpha^2\beta + \alpha^2\gamma + \alpha\beta^2 + \alpha\gamma^2 + \beta^2\gamma + \beta\gamma^2 \\ &\quad + \alpha\beta\gamma + \alpha\gamma\beta + \beta\gamma\alpha) \end{aligned}$$

$$= \sum \alpha^2\beta + 3\alpha\beta\gamma$$

$$\therefore 2 \times 3 = \sum \alpha^2\beta + \frac{3}{2}$$

$$\therefore \sum \alpha^2\beta = 4\frac{1}{2}$$

b) (i) $x - (1-i)$ and $x - (1+i)$ are factors, and so are $x - (2+i)$ and $x - (2-i)$.

$$\begin{aligned} P(x) &= (x - (1-i))(x - (1+i))(x - (2+i))(x - (2-i)) \\ &= (x^2 - 2x + 2)(x^2 - 4x + 5) \end{aligned}$$

$$\text{(ii) } \Delta_1 = 4 - 4 \times 2 < 0$$

$$\Delta_2 = 16 - 4 \times 5 < 0$$

As both quadratics are positive definite, all values of $P(x)$ are greater than or equal to 0.

c) (i) Let $x = \sqrt{y}$

$$\therefore (\sqrt{y})^3 = 6(\sqrt{y})^2 + 7\sqrt{y} - 3 = 0$$

$$\therefore y\sqrt{y} = 6y + 7\sqrt{y} - 3 = 0$$

$$\therefore y\sqrt{y} + 7\sqrt{y} = 6y + 3$$

$$\therefore y^3 + 14y^2 + 49y = 36y^2 + 36y + 9$$

$$\therefore y^3 - 22y^2 + 13y - 9 = 0$$

$$\therefore \text{Equation is } x^3 - 22x^2 + 13x - 9 = 0$$

$$\text{(ii) solution} = \frac{\sum \alpha}{3}$$

$$= \frac{6}{3}$$

$$= 2$$